



Digital Electronics

Special Characteristics Of Digital Logic Families

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INTEGRATED CIRCUITS

- Digital circuits are constructed with integrated circuits.
- An **integrated circuit** (abbreviated IC) is a small silicon semiconductor crystal, called a **chip**, containing the electronic components for the digital gates.
- The various gates are interconnected inside the chip to form the required circuit.
- The chip is mounted in a ceramic or plastic container, and connections are welded to external pins to form the integrated circuit.



Levels of Integration

- Digital ICs are often categorized according to their circuit complexity as measured by the number of logic gates in a single package.
- *Small-scale integration (SSI)* : The number of gates is usually fewer than 10 and is limited by the number of pins available in the IC.
- *Medium-scale integration (MSI)* devices have a complexity of approximately 10 to 100 gates in a single package. They usually perform specific elementary digital operations such as decoders, adders, or multiplexers.
- *Large-scale integration (LSI)* devices contain between 100 and a few thousand gates in a single package. They include digital systems such as processors, memory chips, and programmable logic devices.
- *Very large-scale integration (VLSI)* devices contain thousands of gates within a single package. Examples are large memory arrays and complex microcomputer chips.



Digital Logic Families

- Digital integrated circuits are classified not only by their complexity or logical operation, but also by the specific circuit technology to which they belong. The circuit technology is referred to as a [digital logic family](#).
- The basic circuit in each technology is a NAND, NOR, or an inverter gate.
- The Logic Families are :
 - ❖ RTL → Resistor-Transistor Logic
 - ❖ DTL → Diode Transistor Logic
 - ❖ I²L → Integrated Injection Logic
 - ❖ ECL → Emitter Collector Logic
 - ❖ TTL → Transistor Transistor Logic
 - ❖ MOS → Metal Oxide Semiconductor
 - ❖ CMOS → Complementary MOS



Special Characteristics

- The characteristics of IC digital logic families are usually compared by analyzing the circuit of the basic gate in each family.
- The most important parameters that are evaluated and compared are
 - ❖ Fan-Out
 - ❖ Fan-In
 - ❖ Power Dissipation
 - ❖ Propagation Delay
 - ❖ Speed-Power Product
 - ❖ Noise Margin

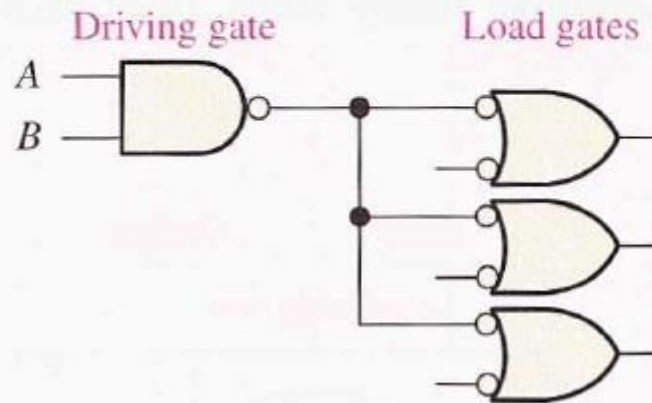


Fan-Out

When the output of a logic gate is connected to one or more inputs of other gates, a load on the driving gate is created, as shown in Figure 1 . There is a limit to the number of load gate inputs that a given gate can drive. This limit is called the **fan-out** of the gate.

► **FIGURE 1**

Loading a gate output with gate inputs.





Fan-Out

- The **fan-out** of a gate specifies the number of standard loads that can be connected to the output of the gate without degrading its normal operation.
- A **standard load** is usually defined as the amount of current needed by an input of another gate in the same logic family. Sometimes the term **loading** is used instead of fan-out.
- This term is derived because the output of a gate can supply a limited amount of current, above which it ceases to operate properly and is said to be overloaded.
- Each input consumes a certain amount of current from the gate output, so that each additional connection adds to the load of the gate.
- Exceeding the specified maximum load may cause a malfunction because the circuit cannot supply the power demanded from it.
- The fan-out is the maximum number of inputs that can be connected to the output of a gate, and is expressed by a number.



Fan-Out

- ❑ The fan-out of the gate is calculated from the ratio I_{OH}/I_{IH} or I_{OL}/I_{IL}
- ❑ For example, the standard TTL gates have the following values for the currents:

$$I_{OH} = 400 \mu\text{A}$$

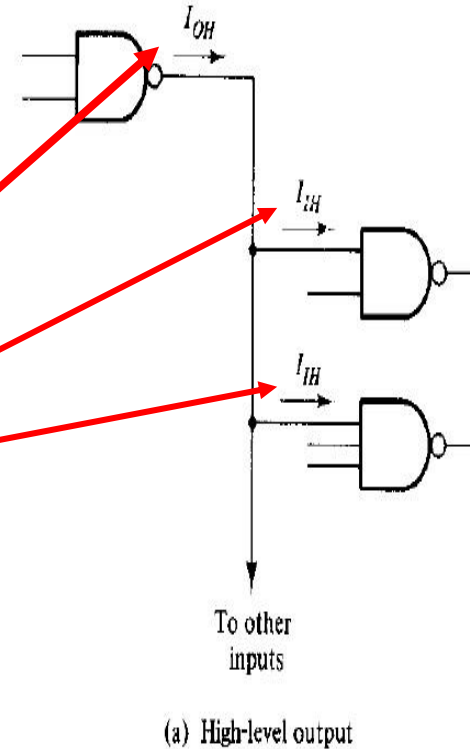
$$I_{IH} = 40 \mu\text{A}$$

$$I_{OL} = 16 \text{ mA}$$

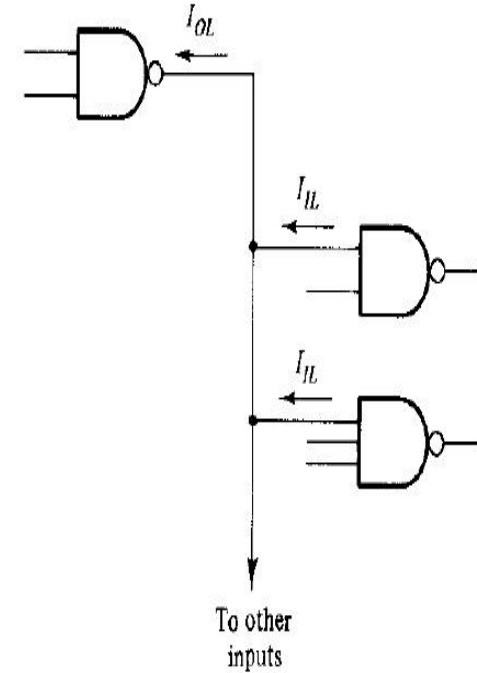
$$I_{IL} = 1.6 \text{ mA}$$

The two ratios give the same number in this case:

$$\frac{400 \mu\text{A}}{40 \mu\text{A}} = \frac{16 \text{ mA}}{1.6 \text{ mA}} = 10$$



(a) High-level output



(b) Low-level output

FIGURE 2

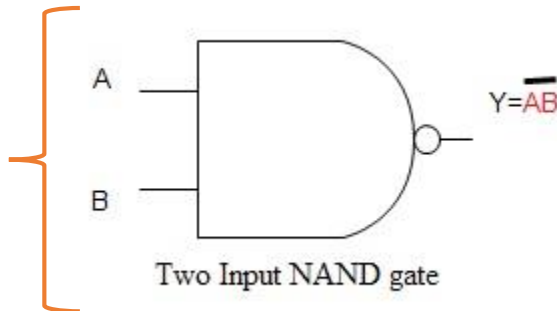
Fan-out computation



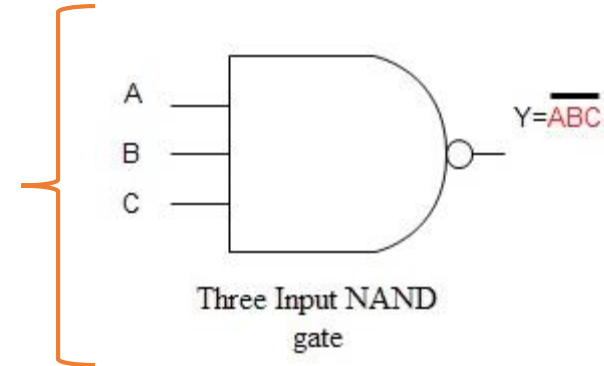
Fan-In

- The fan in defined as the maximum number of inputs that a logic gate can accept.
- If number of input exceeds, the output will be undefined or incorrect.
- It is specified by manufacturer and is provided in the data sheet.

Fan In = 2



Fan In = 3





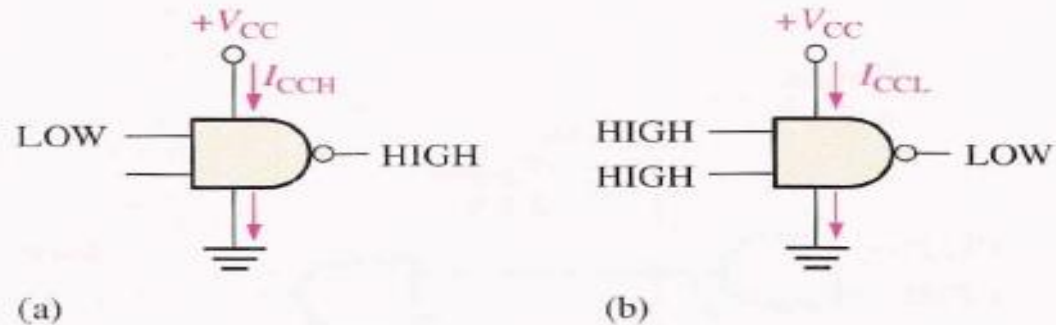
Power Dissipation

Power Dissipation

A logic gate draws current from the dc supply voltage source, as indicated in Figure 3. When the gate is in the HIGH output state, an amount of current designated by I_{CCH} is drawn; and in the LOW output state, a different amount of current, I_{CCL} , is drawn.

FIGURE 3

Currents from the dc supply. Conventional current direction is shown. Electron flow notation is opposite.



As an example, if I_{CCH} is specified as 1.5 mA when V_{CC} is 5 V and if the gate is in a static (nonchanging) HIGH output state, the **power dissipation** (P_D) of the gate is

$$P_D = V_{CC}I_{CCH} = (5 \text{ V})(1.5 \text{ mA}) = 7.5 \text{ mW}$$



Power Dissipation

- The power dissipation is a parameter expressed in milliwatts (mW) and represents the amount of power needed by the gate.
- The number that represents this parameter does not include the power delivered from another gate; rather, it represents the power delivered to the gate from the power supply.
- An IC with four gates will require, from its power supply, four times the power dissipated in each gate.
- When gate is pulsed, its output switches back and forth between HIGH and Low, and the amount of supply current varies between I_{CCH} and I_{CCL} . The average power dissipation depends on the duty cycle is 50%, the output is HIGH half the time and LOW the other half. The average Supply Current is Therefore:

$$I_{CC} = \frac{I_{CCH} + I_{CCL}}{2}$$

The average power dissipation is

$$P_D = V_{CC}I_{CC}$$



Power Dissipation

A certain gate draws $2\ \mu\text{A}$ when its output is HIGH and $3.6\ \mu\text{A}$ when its output is LOW. What is its average power dissipation if V_{CC} is 5 V and the gate is operated on a 50% duty cycle?

Solution The average I_{CC} is

$$I_{CC} = \frac{I_{CCH} + I_{CCL}}{2} = \frac{2.0\ \mu\text{A} + 3.6\ \mu\text{A}}{2} = 2.8\ \mu\text{A}$$

The average power dissipation is

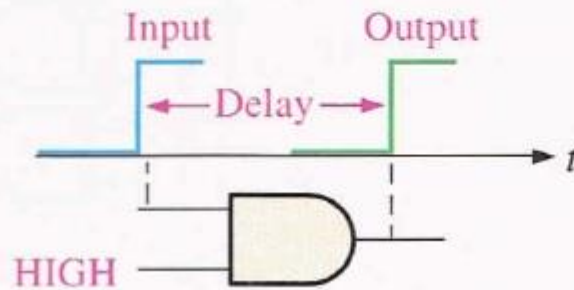
$$P_D = V_{CC}I_{CC} = (5\ \text{V})(2.8\ \mu\text{A}) = \mathbf{14\ \mu\text{W}}$$

Related Problem A certain IC gate has an $I_{CCH} = 1.5\ \mu\text{A}$ and $I_{CCL} = 2.8\ \mu\text{A}$. Determine the average power dissipation for 70% duty cycle operation if V_{CC} is 5 V.



Propagation Delay

When a signal passes (propagates) through a logic circuit, it always experiences a time delay, as illustrated in Figure 4 . . A change in the output level always occurs a short time, called the **propagation delay time**, later than the change in the input level that caused it.



◀ **FIGURE 4**

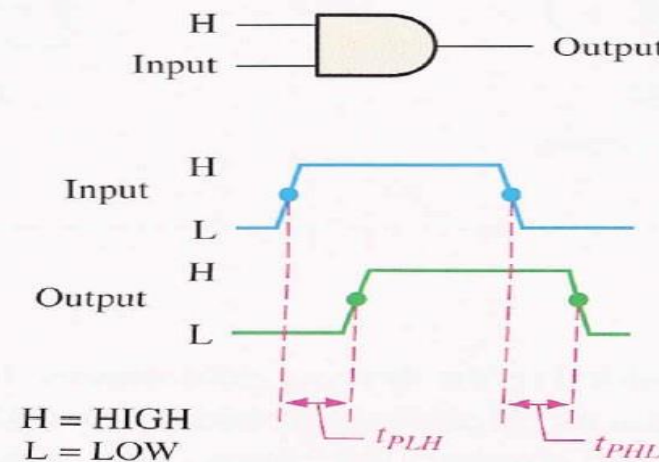
A basic illustration of propagation delay time.



Propagation Delay

- t_{PHL} : The time between a designated point on the input pulse and the corresponding point on the output pulse when the output is changing from HIGH to LOW.
- t_{PLH} : The time between a designated point on the input pulse and the corresponding point on the output pulse when the output is changing from LOW to HIGH.

These propagation delay times are illustrated in Figure 5, with the 50% points on the pulse edges used as references.



◀ **FIGURE 5**
Propagation delay times.



Speed Power Product

- The speed power product provides the basis for the comparison of logic circuits when both propagation delay time and power dissipation are important considerations in the selection of the type of logic to be used in a certain application.
- The lower the speed-power the better.
- The unit of speed-power product is Pico joule(pJ).

$$\text{Speed-Power Product} = \text{Propagation Delay} \times \text{Power Dissipation}$$

□ Table lists parameters for three types of gates. Basing your decision on the speed-power product, which one would you select for best performance?

► **TABLE**

	t_{PLH}	t_{PHL}	P_D
Gate A	1 ns	1.2 ns	15 mW
Gate B	5 ns	4 ns	8 mW
Gate C	10 ns	10 ns	0.5 mW



Noise Margin

- Unwanted signals are referred to as *noise*.
- There are two types of noise to be considered :
 - **DC noise** is caused by a drift in the voltage levels of a signal.
 - AC noise is a random pulse that may be created by other switching signals.
- **Noise margin** is the maximum noise voltage added to an input signal of a digital circuit that does not cause an undesirable change in the circuit output.
- Noise margin is expressed in **volts** and represents the maximum noise signal that can be tolerated by the gate.

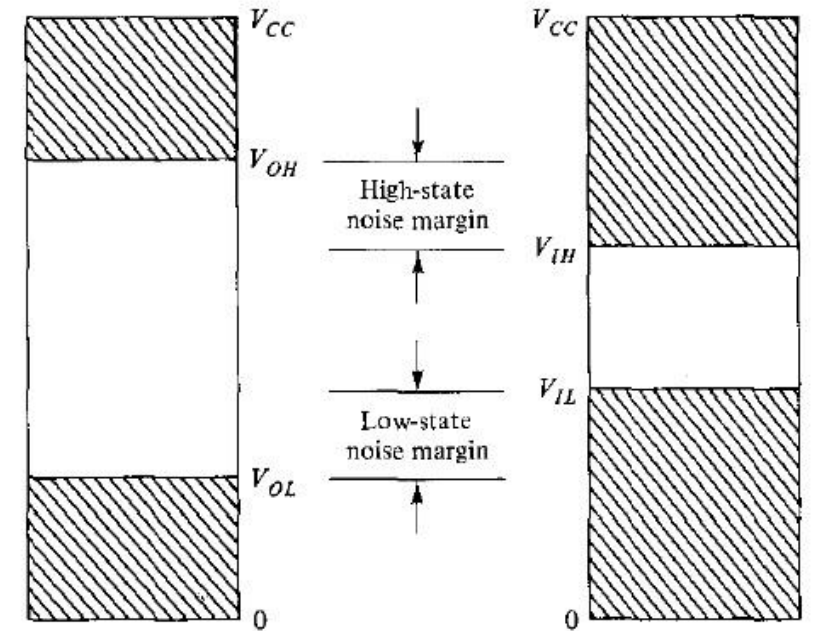


Noise Margin

- There are two values of noise margin specified for a given logic circuit:
 - the HIGH level noise margin (V_{NH}) and
 - the LOW level noise margin (V_{NL}).
- These parameters are defined by the following equations:

$$V_{NH} = V_{OH(\min)} - V_{IH(\min)}$$

$$V_{NL} = V_{IL(\max)} - V_{OL(\max)}$$



(a) Output voltage range

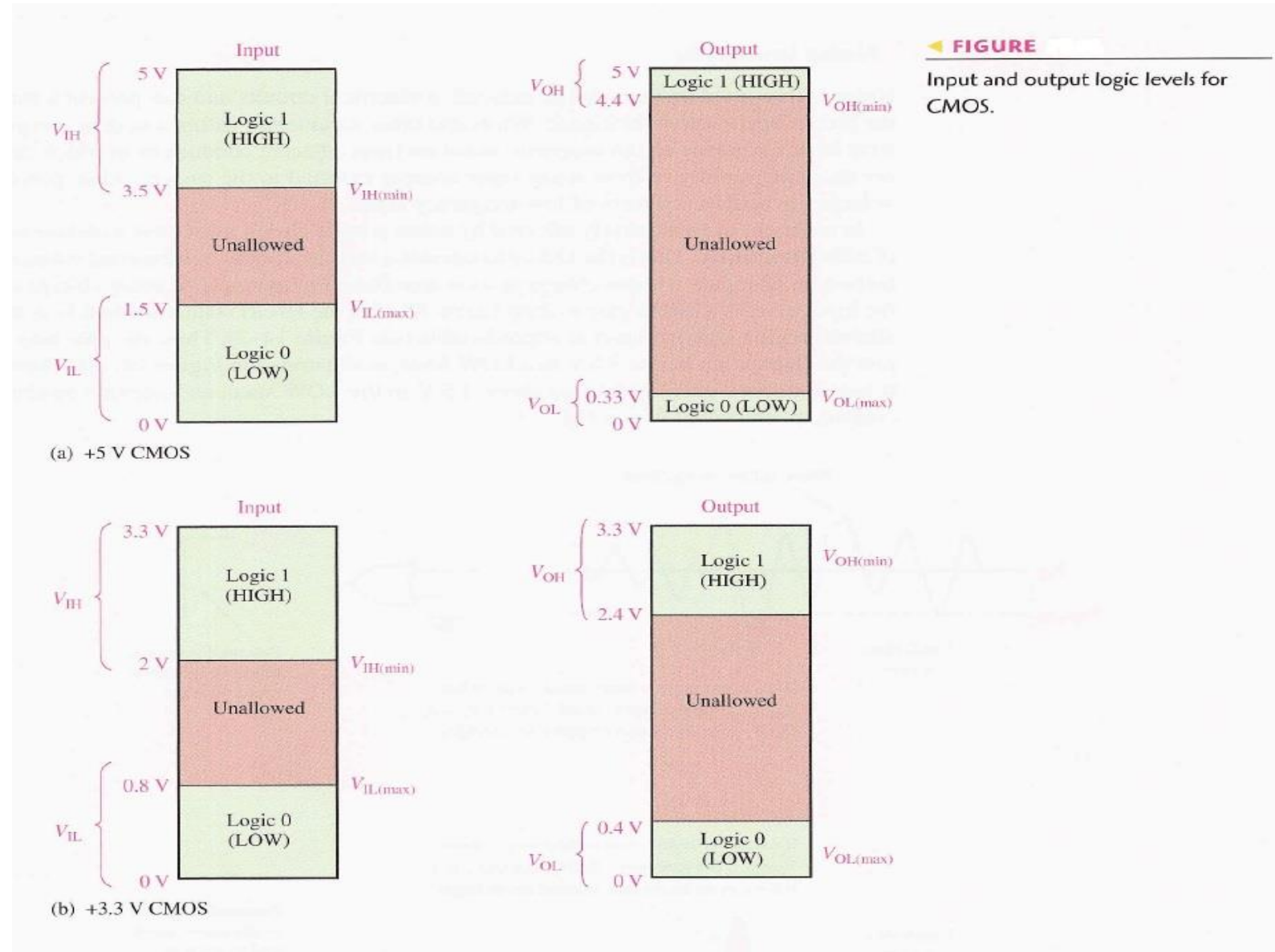
(b) Input voltage range

FIGURE

Signals for evaluating noise margin

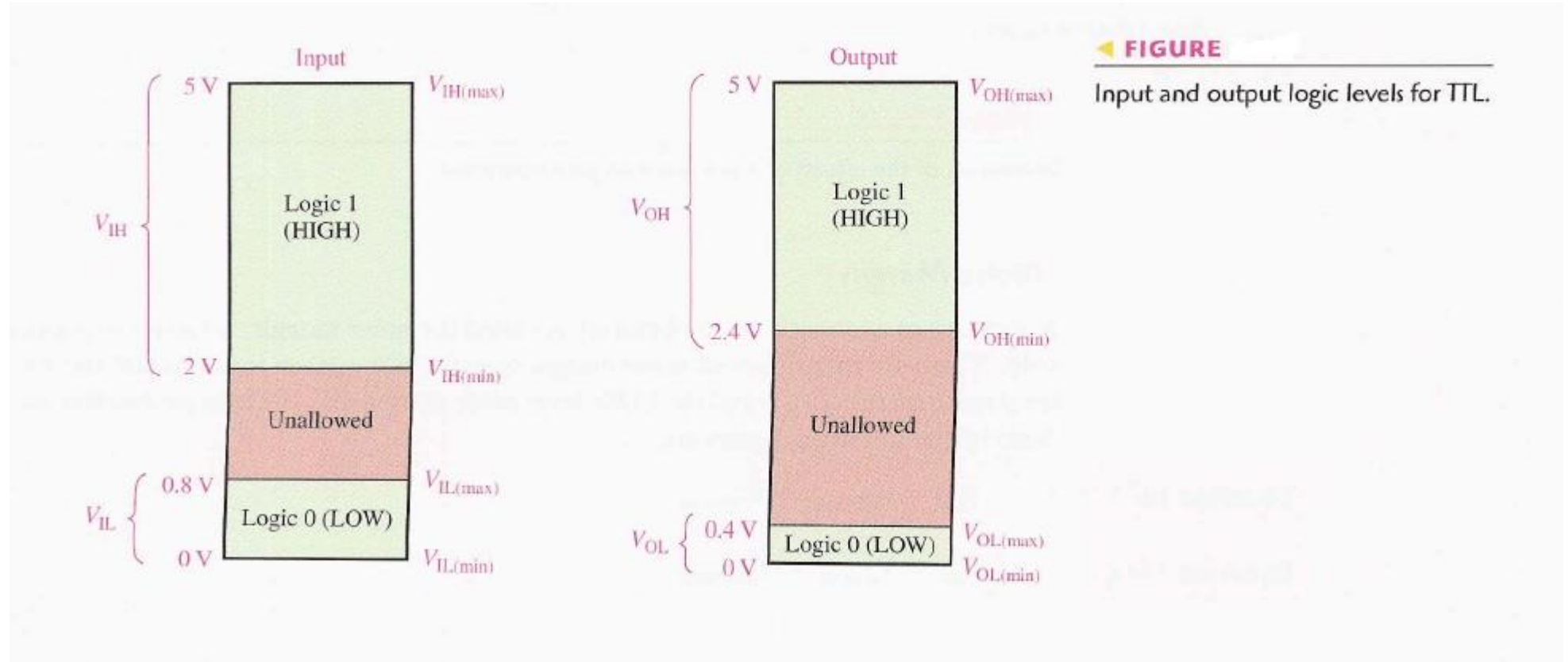


Noise Margin





Noise Margin





Noise Margin

5. Voltage specifications for three types of logic gates are given in Table . Select the gate that you would use in a high-noise industrial environment.

	$V_{OH(MIN)}$	$V_{OL(MAX)}$	$V_{IH(MIN)}$	$V_{IL(MAX)}$
Gate A	2.4 V	0.4 V	2 V	0.8 V
Gate B	3.5 V	0.2 V	2.5 V	0.6 V
Gate C	4.2 V	0.2 V	3.2 V	0.8 V



Bipolar-transistor Characteristics

TABLE 10-1
Typical *npn* Silicon Transistor Parameters

Region	V_{BE} (V)*	V_{CE} (V)	Current Relationship
Cutoff	< 0.6	Open circuit	$I_B = I_C = 0$
Active	$0.6-0.7$	> 0.8	$I_C = h_{FE} I_B$
Saturation	$0.7-0.8$	0.2	$I_B \geq I_{CS}/h_{FE}$

* V_{BE} will be assumed to be 0.7 V if the transistor is conducting, whether in the active or saturation region.



References

1. Thomas L. Floyd, “Digital Fundamentals” 8th edition, Prentice Hall – Pearson Education.
2. M. Morris Mano, “Digital Logic & Computer Design” Prentice Hall